

SOFTWARE ENGINEERING 2

FINAL PROJECT PROPOSAL

Theme: AI-Integrated Information System for Real-World Problems

Student Name: Adrian P. Ebrao

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Subject: Software Engineering 2

Instructor: Mr. Allan Aragon

PROJECT TITLE

TicketSys AI: AI-Integrated Event Ticketing and Attendee Assistance System

I. BACKGROUND OF THE STUDY

Live events such as concerts, sports competitions, conventions, school programs, and community shows continue to grow in number and in audience size. Despite this growth, a large portion of event ticketing in the country is still handled through manual or fragmented methods. Many organizers still sell physical tickets over the counter, accept reservations through social media messages, and record sales in spreadsheets or notebooks. These practices are difficult to monitor and are prone to human error.

This manual setup creates several recurring problems. Physical tickets can be photocopied or duplicated, which allows counterfeit tickets to reach the venue gate. Because sales records are not updated in real time, the same seat can be sold twice, resulting in overbooking and disputes on the event day itself. Entry validation is done by visual inspection, so gate staff must check each ticket by hand. This slows down the entry process and produces long queues outside the venue. Organizers, on the other hand, only discover how well an event performed after the event has ended, since they have no live view of ticket sales, remaining inventory, or revenue.

A second problem, which is often overlooked, is customer communication. Before an event, buyers ask the same questions repeatedly. They ask about the schedule, the venue location, the difference between ticket tiers, the seating arrangement, the refund and reschedule policy, the age restrictions, and the status of their own purchase. Organizers usually answer these questions manually through chat or phone, often outside office hours. This consumes a significant amount of staff time and still leaves many buyers waiting for a reply. Buyers who receive no timely answer frequently abandon the purchase.

Another difficulty is event discovery. As the number of listed events increases, users must scroll through long lists and apply filters manually to find events that match their interests, budget, and schedule. Without a personalized view, relevant events are easily missed, and smaller events receive very little visibility.

An information system can solve the operational side of these problems by centralizing events, ticket inventory, payments, and validation records into one platform. However, a conventional system still cannot respond to a buyer at two o'clock in the morning, cannot explain a policy in the buyer's own words, and cannot suggest events that fit a particular user. This is where Artificial Intelligence becomes necessary. By integrating a Generative AI assistant that is grounded on the platform's own event data and official policies, the system can respond to attendee inquiries instantly, guide users through seat selection and checkout, and recommend events based on the user's history. The AI component supports the information system rather than replacing it. Ticket issuance, payment, inventory control, and entry validation remain the responsibility of the core system.

The proposed system, TicketSys AI, therefore aims to modernize the ticketing process from event discovery to venue entry, while reducing the manual support workload of organizers through AI-assisted communication and recommendation.

II. PROBLEM STATEMENT

The proposed system aims to address the following problems:

- Counterfeit and duplicated tickets allow unauthorized entry and cause revenue loss for organizers.
- The absence of real-time inventory synchronization causes overbooking and double-booking of the same seat.
- Manual ticket inspection at the gate is slow and produces long queues at the venue entrance.
- Attendees wait a long time for answers to repetitive inquiries about schedules, venues, ticket tiers, and policies.
- Organizers spend excessive staff hours responding manually to the same customer questions.
- Users find it difficult to discover events that match their interests because listings are long and unpersonalized.
- Organizers have limited visibility of sales performance while the ticket sale is still ongoing.

III. PROPOSED SYSTEM

The proposed TicketSys AI is a web-based Event Ticketing Management System that handles the complete ticketing cycle, from event discovery to venue entry validation, and adds an AI layer for attendee assistance and event recommendation.

Customers will be able to browse and filter upcoming events, view complete event details, choose a ticket tier or a specific seat through an interactive seating map, pay securely through an integrated payment gateway, and receive a digital ticket that contains a unique QR code. All purchased tickets are stored in the customer account and can be viewed or downloaded at any time.

Event organizers will be able to create and publish events, define ticket types and pricing tiers such as VIP and General Admission, design custom seating layouts using sections, rows, and seats, close ticket sales when capacity is reached, and monitor sales and revenue through an analytics dashboard that updates in real time.

Venue gate staff will be able to scan the QR code of a digital ticket at the entrance. The system verifies the ticket against the database, marks a valid ticket as used, and rejects tickets that are invalid, expired, or previously scanned. This prevents ticket reuse and fraudulent entry.

An integrated AI Ticketing Assistant will operate on top of these modules. It answers attendee questions using the platform's approved event information and policy documents, helps users choose an appropriate ticket tier, retrieves the status of a user's own order, and recommends events that match the user's previous purchases. Inquiries that require a business decision, such as a refund approval, are escalated to the organizer instead of being answered by the AI.

IV. AI INTEGRATION

Primary AI Feature

Generative AI Ticketing Assistant (Retrieval-Augmented Chatbot)

The system will integrate a Generative AI assistant built using a Large Language Model with Retrieval-Augmented Generation (RAG). The assistant is grounded on the platform's own knowledge base, which includes the published event catalog, venue and seating information, ticket tier descriptions, pricing rules, the refund and reschedule policy, entry guidelines, and the frequently asked questions approved by the organizers. When a user asks a question, the system first retrieves the relevant records from this knowledge base and then instructs the language model to answer only from the retrieved content.

The AI assistant will:

- Answer attendee inquiries about event schedules, venues, ticket availability, and prices at any time of the day.
- Explain the difference between ticket tiers and help the user choose a suitable seat or section.
- Explain the refund, reschedule, entry, and age-restriction policies of a specific event.
- Retrieve the status of the logged-in user's own booking, ticket, and payment through secure system functions.
- Guide first-time users through the browsing, seat selection, and checkout process.
- Send reminders about upcoming events and the entry requirements of the venue.
- Escalate the conversation to the organizer or support staff when the request requires human approval.

Supporting AI Features

- **AI Event Recommendation.** A recommendation model analyzes the user's purchase history, viewed events, preferred categories, price range, and location to generate a personalized event feed. This improves event discovery for attendees and increases visibility for smaller events.
- **AI Sales Insight Summary.** The assistant generates a plain-language summary of the organizer's sales dashboard, describing which ticket tiers are selling fastest, which sections remain unsold, and how the current pace compares with previous events.

How the AI Improves the System

The AI feature improves accessibility by allowing attendees to receive accurate answers instantly without waiting for a human representative. It reduces the repetitive support workload of organizers, shortens the time between inquiry and purchase, and improves event discovery through personalized recommendations. The AI supports decision-making and user experience only. It does not issue tickets, does not process payments, does not modify ticket inventory, and does not validate entry. These remain functions of the core information system.

V. TARGET USERS

Primary Users

- Customers / Attendees
- Event Organizers
- Venue / Gate Staff

Secondary Users

- Venue Management
- Marketing and Customer Support Staff
- System Administrator

VI. PROJECT OBJECTIVES

General Objective

To develop an AI-Integrated Event Ticketing Management System that secures the ticketing process from purchase to venue entry, and that improves attendee support and event discovery through an AI assistant.

Specific Objectives

- To digitize the ticket purchasing process from event browsing to checkout.
- To centralize event, ticket, seat, and transaction records in a single database.
- To generate secure digital tickets with unique QR codes to prevent counterfeiting.
- To synchronize ticket and seat inventory in real time to prevent overbooking.

- To provide a fast QR-based entry validation interface for venue gate staff.
- To provide AI-assisted attendee support through a chatbot grounded on approved event information.
- To provide personalized event recommendations based on user activity and purchase history.
- To generate sales, attendance, and revenue reports automatically for organizers.

VII. SCOPE

The proposed system will provide secure role-based login for customers, organizers, and gate staff. It will maintain event records, ticket types, and seating layouts; allow browsing, searching, and filtering of events; support interactive seat selection with temporary reservation holds; process payments through an integrated payment gateway in sandbox mode; generate and deliver digital tickets with unique QR codes; maintain purchase history; synchronize ticket inventory in real time; validate tickets at the gate through QR scanning; generate sales and attendance reports; and provide AI-powered attendee assistance and event recommendations through a chatbot interface.

VIII. LIMITATIONS

The proposed system will not replace human organizers or venue staff. Final decisions on refunds, cancellations, event approval, and entry disputes remain the responsibility of authorized personnel. The AI assistant will respond only on the basis of approved event data and official platform documents, and it will not provide legal, medical, or financial advice. The AI assistant cannot issue, modify, or cancel a ticket by itself; it can only guide the user or forward the request to the responsible staff.

Payment processing will be implemented using the Maya Sandbox environment, so no actual funds will be transferred during the project. The system will not handle physical or over-the-counter ticket printing, and it will not manage venue logistics such as staffing, catering, or equipment. Offline operation is not supported; both the ticketing functions and the AI assistant require an internet connection.

IX. PROPOSED SYSTEM FEATURES

- User Authentication and Role Management
- Event Browsing, Search, and Filtering
- Event Details and Availability Display
- Interactive Seat Selection and Seating Map
- Ticket Reservation and Checkout with Temporary Hold
- Secure Payment Processing through a Payment Gateway
- Digital Ticket Generation with Unique QR Code

- Ticket Delivery and Purchase History
- Real-Time Ticket Inventory Synchronization
- QR Code Scanning and Entry Validation
- Organizer Event, Pricing, and Seating Layout Management
- Sales and Analytics Dashboard
- AI Ticketing Assistant (Chatbot)
- AI Event Recommendation Feed
- Notification and Announcement Module
- Report Generation

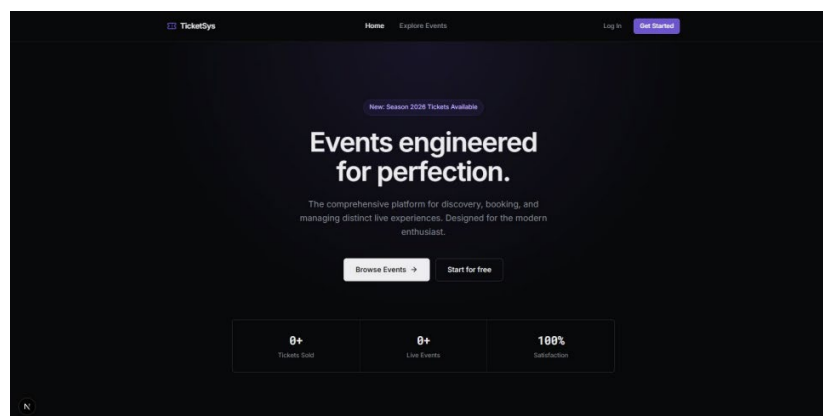
X. EXPECTED BENEFITS

- Reduces ticket fraud through unique QR-coded digital tickets and single-use validation.
- Prevents overbooking and double-booking through real-time inventory synchronization.
- Shortens entry queues by replacing manual inspection with QR scanning.
- Provides attendees with instant and consistent answers at any time through the AI assistant.
- Reduces the repetitive customer-support workload of organizers and support staff.
- Improves event discovery and ticket conversion through personalized recommendations.
- Gives organizers live visibility of sales, revenue, and remaining inventory for better decisions.
- Centralizes ticketing records, which improves data accuracy and simplifies reporting.

XI. INITIAL USER INTERFACE

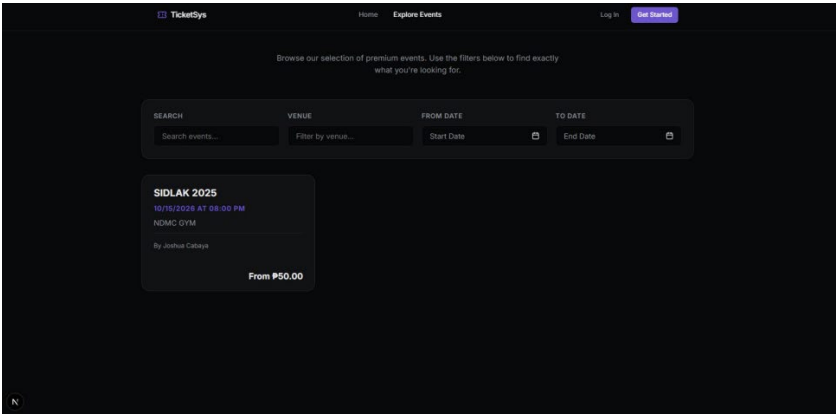
Screen 1

Home Page



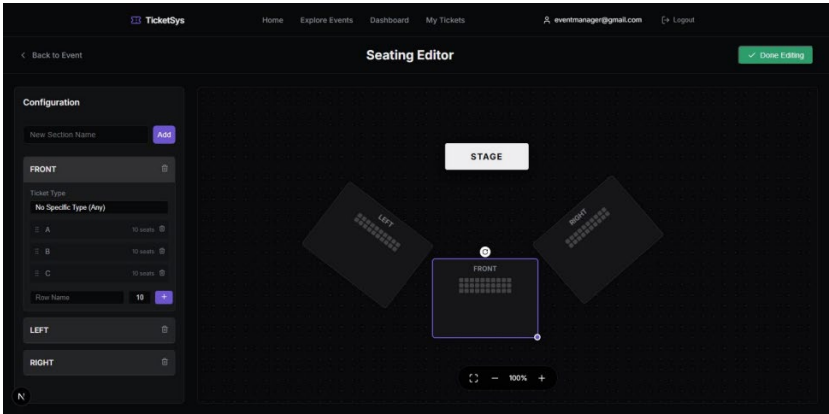
Screen 2

Event Listing



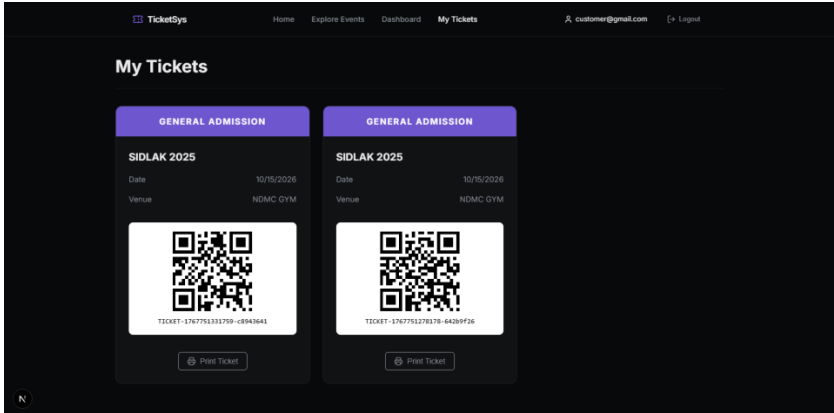
Screen 3

Event Organizer Seating Editor



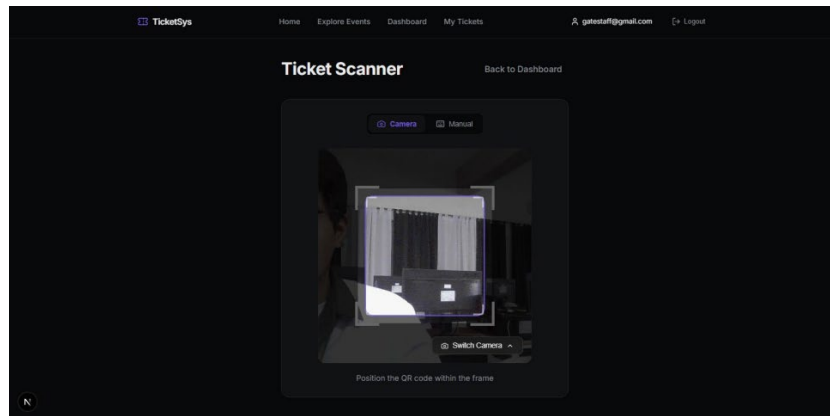
Screen 4

Customer My Tickets Page



Screen 5

Gate Staff QR Scanning and Validation Screen



XII. TECHNOLOGY STACK

Frontend:

Next.js (React) with Tailwind CSS

Backend:

ASP.NET Core Web API

Database:

PostgreSQL, with pgvector for the AI knowledge base embeddings

AI Integration:

Large Language Model API (OpenAI or Anthropic Claude) using a Retrieval-Augmented Generation pipeline over institutional and event documents; a content-based recommendation model for the personalized event feed

Payment Gateway:

Maya Sandbox

QR Code Generation and Scanning:

QRCode for generation and a browser-based scanner library for gate validation

Hosting:

Vercel (Frontend) and Render or Azure (Backend and Database)